

# MATH and TRIGONOMETRY functions, Referencing, Naming Ranges.

① How do you use the SUMIF functions to add values that meet a specific criterion?

→ The SUMIF function is used to add values in a range that satisfy a given condition. It evaluates each cell in a specified range and sums the corresponding values if the condition is met.

Syntax:  $=\text{SUMIF}(\text{range}, \text{criteria}, [\text{sum\_range}])$

• range → cells to test

• criteria → condition to meet

• sum\_range → cells to add

Example.

$=\text{SUMIF}(A1:A10, "Apple", B1:B10)$  adds values in B1:B10 where A1:A10 contains "Apple".

② What is the difference between the SUM and SUMPRODUCT function?

→ The SUM functions simply adds numbers in a range.

The SUMPRODUCT functions multiplies corresponding elements in arrays and then sum the results.

SUM syntax:  $=\text{SUM}(A1:A5)$

SUMPRODUCT Syntax:

$=\text{SUMPRODUCT}(A1:A5, B1:B5)$

SUMPRODUCT is useful for weighted calculations and Conditional totals without helper columns.

③ How would you use the Round, Roundup, Rounddown functions?

- These functions control numerical rounding
- ROUND rounds to the nearest value
  - ROUNDUP always rounds away from zero
  - ROUNDDOWN always rounds toward zero.
- Syntax = Round (Number, digits)

Example:

- = ROUND (12.456, 2)  $\rightarrow$  12.46
- = ROUNDUP (12.451, 2)  $\rightarrow$  12.46
- = ROUNDDOWN (12.459, 2)  $\rightarrow$  12.45

④ Explain how to use the Countif functions to Count cells that meet a condition

→ The Countif functions counts the number of cells that meet a specific condition.

Syntax := Countif (range, criteria)

It is commonly used for counting text, number, or logical conditions.

Example = COUNTIF (A1:A20, ">50") Counts cells with values greater than 50.

⑤ What is the purpose of the PI functions in Excel?

→ The PI function returns the mathematical constant  $\pi$  (pi) approximately 3.141592654.

It is mainly used in trigonometric and geometric calculations involving circles.

Syntax :  $=PI(C)$

Example :  $=PI(C)*A1/2$  calculates the ~~the~~ area of a circle where  $A1$  is the radius.

⑥ How do you apply the SIN, COS, and TAN functions in excel?

→ The SIN, COS, and TAN functions calculate trigonometric values of angles given in radians.

Syntax

$=SIN(\text{number})$

$=COS(\text{number})$

$=TAN(\text{number})$

If the angle is in degrees, use ~~Radians~~ RADIANS().

Example  $=SIN(RADIANS(30))$  returns 0.5.

⑦ How can naming ranges improve the readability and management of formulas?

→ Naming ranges assigns meaningful names to all ranges, making formulas easier to understand and manage.

Named ranges reduce error and simplify complex formulas.

Example : Instead of  $=SUM(A1:A10)$ , used  $=SUM(Sales)$

Named ranges also update automatically when the range size changes.

8) what are dynamic named ranges, and how do you create one?

Dynamic named ranges automatically expand or contract when data is added or removed. They are created using functions like `OFFSET` or `INDEX`.

Example.

`=OFFSET(A1, 0, 0, COUNTA(A:A), 1)`

Dynamic ranges are useful for charts and dashboards that update automatically.

How do you reference cells from another worksheet in a formula?

9) To reference cells from another worksheet, use the sheet name followed by an exclamation mark.

Syntax = SheetName!Cell Reference.

Example: `=Sheet2!A1`

For sheet names with spaces: `= 'Sales Data'!B2`.

10) what is an array formula, and how does it differ from regular formula?

→ An array formula performs calculations on multiple values simultaneously. Unlike regular formulas, it can return multiple results or a single aggregated result.

Older versions require `Ctrl + Shift + Enter`, while newer Excel versions support dynamic arrays.

Example: `=SUM(A1:A5 * B1:B5)` multiplies and sums arrays in one step.